Total No. of Questions : 4]

SEAT No. :

PA-10285

[Total No. of Pages : 2

[6010]-70
B.E. (E & TC) (Insem)
FIBER OPTIC COMMUNICATION
(2019 Pattern) (Semester - VIII) (404190)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates :

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Using simple ray theory, describe the mechanism for the transmission of light within an optical fiber. With the help of suitable diagram discuss what is meant by the acceptance angle and numerical aperture of the fiber. **[5]**

b) A manufacturer wishes to make a silica-core, step-index fiber with V = 75 and a numerical aperture NA = 0.30 to be used at 820 nm. If n1 = 1.458. What should the core size and cladding refractive index be? **[5]**

c) Describe with the aid of simple ray diagrams. the MMSI and SMSI fiber. Compare advantages and disadvantages of these two types of fibers. **[5]**

OR

Q2) a) Explain the key elements of the optical fiber communication system with neat diagram and identify the three transmission windows on the fiber attenuation curve. **[6]**

b) The mean optical power launched into an optical fiber link is 1,5 mW and the fiber has an attenuation of 0.5dB/Km. Determine the maximum possible link length without repeaters when the minimum mean optical power level required at the detector is 2μW. **[4]**

c) Classify and explain the various loss mechanisms in optical fibers. **[5]**

P.T.O.

Q3) a) The radiative and nonradiative recombination lifetimes of the minority carriers in the active region of a double heterojunction LED are 60 ns and 100 ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 μm at a drive current of 40 mA. **[6]**

b) Describe with the help of suitable diagram the mechanism giving the emission of light from an LED. State useful properties of LED to be used in optical fiber communication. **[5]**

c) State the semiconductor materials used for various optical sources. **[4]**

OR

Q4) a) Explain the following terms **[5]**

i) Absorption

ii) Spontaneous Emission

iii) Stimulated Emission

b) Compare LED with LASER w.r.t. following points: **[5]**

i) Output Power

ii) Spectral width

iii) Coupled Power

iv) E/O efficiency

v) Speed

c) A GaAs laser operating at 900nm has a 300μm length and refractive index n=4.3. Find the following **[5]**

i) Frequency spacing

ii) Wavelength spacing

iii) Number of modes

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-> a) Mechanism for the Transmission of Light within an Optical Fiber Using Simple Ray Theory :

Light exhibits both wave and particle nature. In optical fibers, the transmission of light is primarily explained using ray theory, which considers light as traveling in straight-line paths, called rays.

An optical fiber consists of two main parts:

- Core: The inner part where light travels.

- Cladding: The outer layer that surrounds the core and has a lower refractive index.

- When a light ray enters the fiber at a certain angle, it bends due to refraction at the core-cladding interface.

- If the angle of incidence is greater than a critical angle, the ray is totally internally reflected and continues bouncing within the core.

- This process repeats throughout the length of the fiber, enabling the transmission of light from one end to the other.

Acceptance Angle and Numerical Aperture of Optical Fiber

Acceptance Angle:

- The acceptance angle is the maximum angle at which light can enter the fiber and still undergo total internal reflection within the core.
- If the light enters at an angle greater than the acceptance angle, it will not reflect properly and will be lost.

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- The acceptance angle determines the amount of light that can be coupled into the fiber.

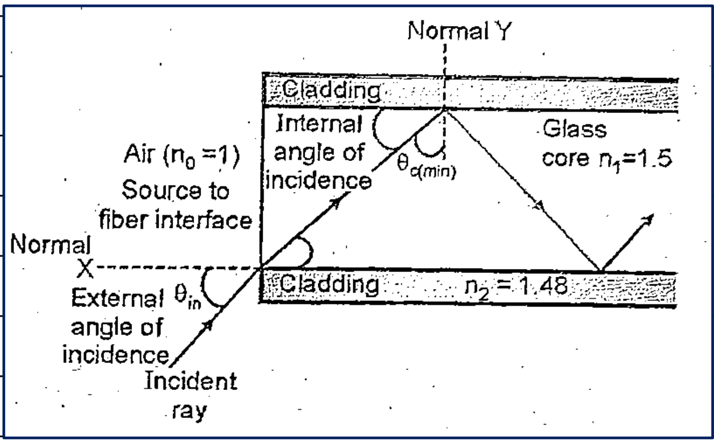

Figure: Acceptance Angle in Optical Fiber

Numerical Aperture (NA):

- The Numerical Aperture (NA) is a dimensionless number that indicates the light-gathering ability of an optical fiber.
- It is defined as:

NA = sin(acceptance angle)

- A higher NA allows more light to enter the fiber, improving signal transmission efficiency.
- The value of NA depends on the refractive indices of the core and cladding:

$$NA = \text{sqrt}(n1^2 - n2^2)$$

Where:
- n1 = Refractive index of the core
- n2 = Refractive index of the cladding

Significance of NA:

- Determines how much light can enter the fiber.

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- Affects the fiber's ability to capture and transmit signals efficiently.
- Higher NA fibers are used in applications requiring high light collection efficiency.

-> b) Given Data:

- Normalized Frequency (V): 75
- Numerical Aperture (NA): 0.30
- Operating Wavelength (λ): 820 nm = 820×10^{-9} m
- Core Refractive Index (n_1): 1.458

Step 1: Calculate Core Radius (a):

- Formula: $V = (2\pi a/\lambda) \times NA \rightarrow a = (V \times \lambda) / (2\pi \times NA)$
- Substitution:
  - $a = (75 \times 820 \times 10^{-9}\ m) / (2 \times 3.1416 \times 0.30)$
  - $a \approx 32.58 \times 10^{-6}\ m = 32.58\ \mu m$
- Core Diameter:
  - $2 \times a = 2 \times 32.58\ \mu m = 65.16\ \mu m$

Step 2: Calculate Cladding Refractive Index (n_2):

- Formula: $NA = sqrt(n_1^2 - n_2^2) \rightarrow n_2 = sqrt(n_1^2 - NA^2)$
- Substitution:
  - $n_2 = sqrt(1.458^2 - 0.30^2) = sqrt(2.126 - 0.09) = sqrt(2.036)$
  - $n_2 \approx 1.427$

Final Results:

- Core Diameter: 65.16 μm
- Cladding Refractive Index (n_2): 1.427

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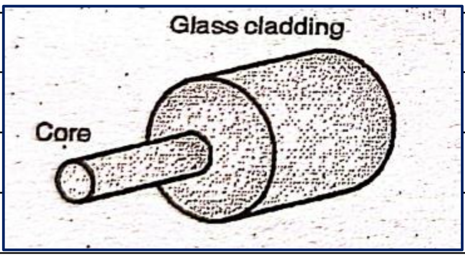
-> c) MMSI & SMSI :

- Optical fibers are classified into different types based on the mode of propagation and refractive index profile.
- Among these, the two major types of step index fibers are Single Mode Step Index (SMSI) fibers and Multimode Step Index (MMSI) fibers.
- These fibers are widely used in optical communication systems due to their ability to transmit light signals over long distances with minimal loss.

Single Mode Step Index (SMSI) Fiber:

- The Single Mode Step Index (SMSI) fiber is an optical fiber that allows only one mode of light to propagate.
- It has a very small core diameter, which significantly reduces the dispersion of light and improves signal quality over long distances.

Construction:



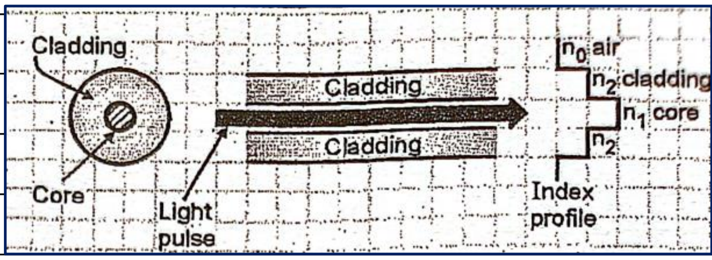


Figure : Single Mode Step Index Fiber Construction

The SMSI fiber consists of:
- Core: A very thin core with a diameter of about 8-10 microns.
- Cladding: A surrounding layer with a lower refractive index than the core.
- Refractive Index Profile: The core has a uniform refractive index (n1), and the cladding has a lower refractive index (n2). The transition between the core and cladding is abrupt, forming a step-like profile.

Light Propagation:

- Since only one mode propagates, the signal remains stable without much dispersion.
- The light travels in a straight line, bouncing minimally off the core-cladding interface.
- This results in low attenuation and high bandwidth.

Multimode Step Index (MMSI) Fiber:

The Multimode Step Index (MMSI) fiber allows multiple modes of light to propagate through the core, leading to higher dispersion compared to SMSI fibers.

Construction:

- Core: A larger core diameter (50-100 microns) compared to SMSI fiber.
- Cladding: Surrounds the core and has a lower refractive index.
- Refractive Index Profile: Similar to SMSI fiber, the refractive index of the core is uniform and higher than the cladding.

Diagram:

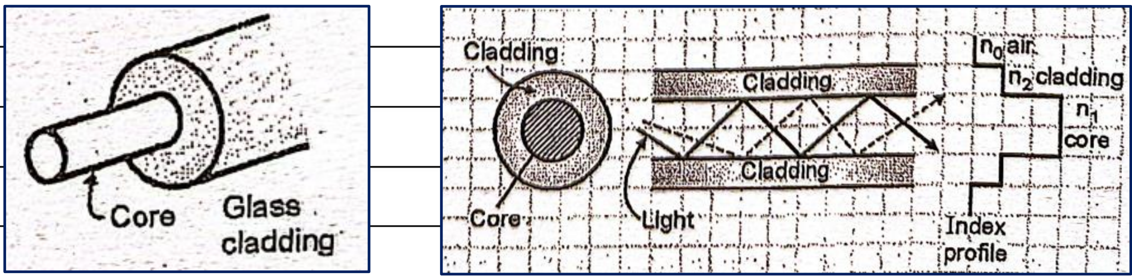


Figure: Multimode Step Index Fiber Construction

Light Propagation:

- Multiple light rays travel through different paths, leading to modal dispersion.

- Signals arrive at different times due to varying path lengths, which can cause signal distortion.
- Suitable for short-distance applications where dispersion is less of a concern.

Advantages of MMSI Fiber:
- Easier to manufacture and lower cost.
- Works well with cheaper LED light sources.
- Ideal for short-distance communication like LANs and CCTV.

Comparison of SMSI and MMSI

Sr. No.	Parameter for Comparison	SMSI Fiber	MMSI Fiber
1.	Number of paths followed by light rays inside fiber	One	Many
2.	Index profile	Step profile (Abrupt change in refractive index at core-cladding interface)	Step profile (Abrupt change in refractive index at core-cladding interface)
3.	Distortion	Less than MMSI fiber	More than SMSI fiber
4.	Source to fiber aperture	Small	Large
5.	Ease of coupling light into the fiber	Difficult	Easy
6.	Bandwidth	High	Low
7.	Data rate	Higher than MMSI	Lower than SMSI
8.	Applications	Wideband long-distance communication	Narrowband short-distance communication
9.	Cost	Expensive	Less expensive
10.	Ease of manufacturing	Difficult	Easy
11.	Phenomenon responsible for light propagation	Reflection	Reflection
12.	Shape of light ray paths	Straight	Straight

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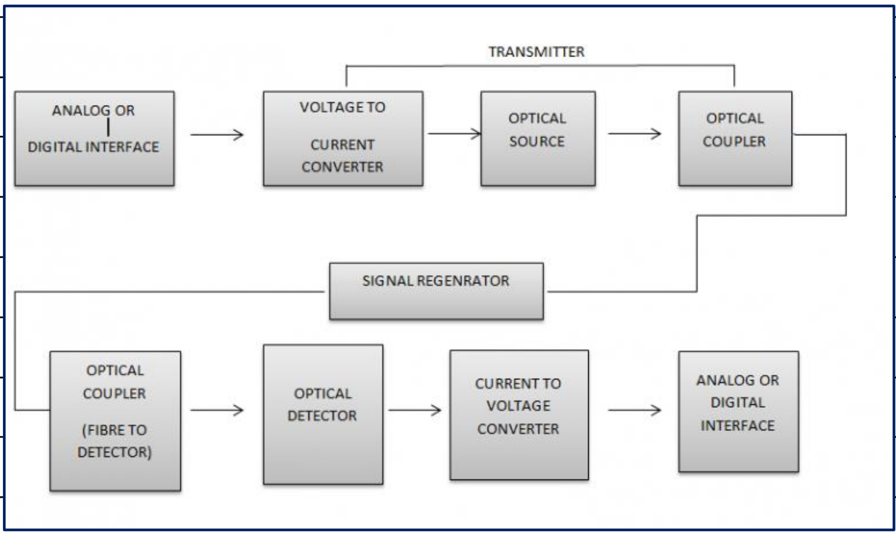
-> a) Optical Fiber Communication

Definition:

- Optical fiber communication is a method of transmitting information from one place to another by sending pulses of light through an optical fiber.
- The light forms an electromagnetic carrier wave that is modulated to carry data.
- Optical fiber communication is widely used in telecommunications and computer networking because of its high speed, bandwidth, and efficiency.

Block Diagram of Optical Fiber Communication System



The optical fiber communication system consists of the following components:

1. Information Source:

- The information source generates the message signal that needs to be transmitted.
- The message can be in analog or digital form, such as voice, video, or data signals.
- The output of the information source is in the form of an electrical signal.

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2. Transmitter:

- The transmitter converts the electrical signal from the information source into an optical signal.
- It consists of:
  - Voltage to Current (V-I) Converter: Converts voltage signals into current signals.
  - Light Source: Converts electrical signals into light signals using infrared LED (IRLED) or Injection Laser Diode (ILD).
  - Optical Coupler (Source to Fiber Interface): Couples the light efficiently into the optical fiber.

3. Optical Fiber:

- A medium used for transmitting the light signal from the transmitter to the receiver.
- It consists of:
  - Core: The central part through which light propagates.
  - Cladding: Surrounds the core and helps in total internal reflection.
  - Buffer Coating: Protects the fiber from mechanical damage and environmental factors.

4. Signal Regenerator (Repeater):

- Optical signals suffer from attenuation and distortion while traveling through fiber.
- A signal regenerator is used to amplify and reconstruct the optical signal before it reaches the receiver.
- It enhances signal strength and reduces noise.

5. Receiver:

- The receiver converts the optical signal back into an electrical signal.
- It consists of:

- Fiber to Detector Interface: Couples light into the detector.
- Light Detector: Converts optical signals into electrical signals (e.g., PIN diode or Avalanche Photodiode).
- Current to Voltage (I-V) Converter: Converts the output current of the detector into a voltage signal.
- The recovered electrical signal is then sent to the destination.

6. Optical Fiber Cable:
   - Optical fibers are packed inside a protective cable.
   - Cables can be installed underground, underwater, or aerially.

Three Optical Transmission Windows

First Transmission Window
- Wavelength Range: 800 nm - 900 nm
- Operating Wavelength: 850 nm
- Characteristics:
  - The first optical window operates in the near-infrared region.
  - Initially, this window was widely used due to the availability of inexpensive light sources such as light-emitting diodes (LEDs) and semiconductor lasers.
  - It is mostly used in short-distance optical fiber communication.
  - The attenuation in this window is relatively high compared to the other two windows, limiting its use for long-distance communication.
  - Multimode fiber is commonly used in this window.

Second Transmission Window
- Wavelength Range: 1260 nm - 1360 nm
- Operating Wavelength: 1310 nm
- Characteristics:

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- The second optical window offers lower attenuation compared to the first window.
  - It became more popular in fiber optic communication due to its zero dispersion characteristics, meaning minimal signal distortion over long distances.
  - This window is used in single-mode fiber transmission, which allows for higher data rates and longer transmission distances.
  - Optical fiber components such as lasers and photodetectors are optimized to work efficiently at this wavelength.

Third Transmission Window

- Wavelength Range: 1500 nm - 1600 nm
- Operating Wavelength: 1550 nm
- Characteristics:
  - The third optical window is considered the most important for modern optical communication systems.
  - It offers the lowest attenuation, making it ideal for long-distance and high-speed optical communication.
  - This window is widely used in fiber optic networks, including undersea communication cables.
  - Optical amplifiers, such as erbium-doped fiber amplifiers (EDFAs), work efficiently in this window, allowing signal regeneration over long distances without electrical conversion.

-> b) Given Data:

- Initial Optical Power (P_0): 1.5 mW = 1.5×10^{-3} W
- Fiber Attenuation (α): 0.5 dB/km
- Minimum Optical Power Required (Pmin): 2 μW = 2×10^{-6} W

Step 1: Calculate Total Allowable Power Loss (A):

- Formula: Power Loss (dB) = $10 \times \log_{10}(P_0 / P_{min})$
- Substitution:
  - $A = 10 \times \log_{10}(1.5 \times 10^{-3}\ W / 2 \times 10^{-6}\ W)$
  - $A = 10 \times \log_{10}(750)$
  - $\log_{10}(750) \approx 2.875 \rightarrow A = 10 \times 2.875 = 28.75$ dB

Step 2: Calculate Maximum Link Length (L):

- Formula: Total Attenuation (A) = Attenuation per km (α) × Length (L)
- Rearranged Formula: $L = A / \alpha$
- Substitution:
  - $L = 28.75$ dB / 0.5 dB/km = 57.5 km

Final Result:

- Maximum Link Length Without Repeaters: 57.5 km

-> c) Various Loss Mechanisms in Optical Fibers :

- Optical fiber communication offers numerous advantages over traditional communication systems.
- However, one of its significant challenges is power loss, also known as attenuation.
- This loss results in a reduction of signal power as it travels through the fiber, affecting system performance in various ways.

Effects of Losses in Optical Fiber

- Reduction in System Bandwidth: Attenuation limits the bandwidth, thereby reducing the amount of data that can be transmitted.

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- Reduction in Transmission Rate: A weaker signal leads to slower data transmission.
- Reduced Efficiency: Increased loss requires higher amplification, reducing overall efficiency.
- Decreased System Capacity: Higher losses limit the amount of information that can be sent over long distances.

Classification of Losses in Optical Fiber

Below is a classification of various types of losses encountered in optical fiber communication:

Losses Due to Attenuation

1. Absorption Loss
2. Scattering Loss
3. Radiation Loss
4. Coupling Loss
5. Dispersion Loss

1. Attenuation in Optical Fiber

Definition:

Attenuation is the reduction in signal power as it travels through the optical fiber. It is caused by various factors such as absorption, scattering, and bending of light within the fiber.

Types of Attenuation Losses

- Intrinsic Losses: These occur due to the fundamental properties of the fiber material. The interaction of light with the core material leads to scattering and absorption.
- Fresnel Losses: These occur due to sudden changes in refractive index, causing reflection and bending of light rays.
- Rayleigh Scattering Losses: The microscopic variations in fiber structure cause light to scatter, leading to signal loss.

2. Absorption Losses

Definition:
Absorption losses occur due to impurities or defects in the fiber material, which absorb the transmitted light and convert it into heat.

Causes of Absorption Loss
- Intrinsic Absorption: Caused by the fundamental atomic structure of the fiber material.
- Extrinsic Absorption: Caused by impurity atoms such as iron, copper, and hydroxyl ions in the fiber.
- Atomic Defects: Imperfections in the glass structure result in additional absorption.

3. Scattering Losses

Definition:
Scattering losses occur due to the microscopic variations in the fiber material, which cause light to scatter in different directions.

Types of Scattering Losses
- Linear Scattering Losses: These occur due to small structural variations in the fiber.

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- Non-Linear Scattering Losses: These happen at high power levels and include Stimulated Brillouin Scattering (SBS) and Stimulated Raman Scattering (SRS).

4. Coupling Losses

Definition:
Coupling losses occur due to imperfect alignment between fiber connections, leading to power loss at joints or connectors.

Types of Coupling Losses
- Lateral Misalignment: Occurs when two fibers are not perfectly aligned.
- Gap Misalignment: A physical gap between two fiber ends causes power loss.
- Angular Misalignment: Misalignment of fiber angles leads to inefficient light coupling.
- Imperfect Surface Finish: Rough surfaces cause scattering and power loss.

5. Dispersion Losses

Definition:
Dispersion is the spreading of an optical pulse as it travels through the fiber, leading to signal distortion.

Types of Dispersion
- Modal Dispersion: Different modes travel at different speeds, causing pulse broadening.
- Chromatic Dispersion: Different wavelengths travel at different speeds, affecting signal clarity.
- Polarization Mode Dispersion: Variations in fiber material affect polarization states, leading to additional loss.

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6. Fiber Bending Losses

Definition:

Fiber bending losses occur when an optical fiber is bent at an excessive angle, causing light to escape from the core.

Types of Bending Losses

- Macrobending Losses: Large bends in the fiber cause leakage of light.

- Microbending Losses: Tiny deformations in the fiber lead to scattering and loss.

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-> a) Formula for Total Carrier Recombination Lifetime

To calculate the total carrier recombination lifetime (τ_{total}), we use the relation:

$$(1 / \tau_{total} = 1 / \tau_r + 1 / \tau_{nr})$$

where:

- τ_r = 60 nanoseconds (ns)
- τ_{nr} = 100 nanoseconds (ns)

Now, substituting the given values:

$1 / \tau_{total} = (1 / 60) + (1 / 100)$

$1 / \tau_{total} = (100 + 60) / (60 \times 100)$

$1 / \tau_{total} = 160 / 6000$

$\tau_{total} = 6000 / 160$

$\tau_{total} = 37.5$ ns

Thus, the total carrier recombination lifetime is 37.5 nanoseconds.

Calculation of Internally Generated Power

The power internally generated within the LED is given by:

$P_{internal} = (q \times I \times \lambda) / (h \times c)$

where:

- q = charge of an electron = 1.6×10^{-19} C
- I = drive current = 40 mA = 40×10^{-3} A
- λ = emission wavelength = 0.87 µm = 0.87×10^{-6} m

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- h = Planck's constant = 6.626×10^{-34} J·s
- c = speed of light = 3×10^8 m/s

Now, substituting the given values:

$P_{internal} = (1.6 \times 10^{-19} \times 40 \times 10^{-3} \times 0.87 \times 10^{-6}) / (6.626 \times 10^{-34} \times 3 \times 10^8)$

$P_{internal} = (5.568 \times 10^{-27}) / (1.9878 \times 10^{-25})$

$P_{internal} = 0.028$ W or 28 mW

Thus, the internally generated power within the LED is 28 mW.

-> b) Light Emitting Diode (LED) and Its Working in Optical Fiber Communication

- A Light Emitting Diode (LED) is a special type of p-n junction diode that emits light when it is forward biased.
- It is widely used in optical fiber communication, display panels, and indicator lights due to its high efficiency and long lifespan.
- The light emission occurs due to the recombination of charge carriers, which releases energy in the form of photons.

Structure of LED

- An LED consists of two terminals: Anode and Cathode.
- The Anode is connected to the p-type semiconductor (which contains an excess of holes).
- The Cathode is connected to the n-type semiconductor (which contains an excess of electrons).
- A depletion region is formed at the junction due to the recombination of holes and electrons.

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Working Mechanism of LED

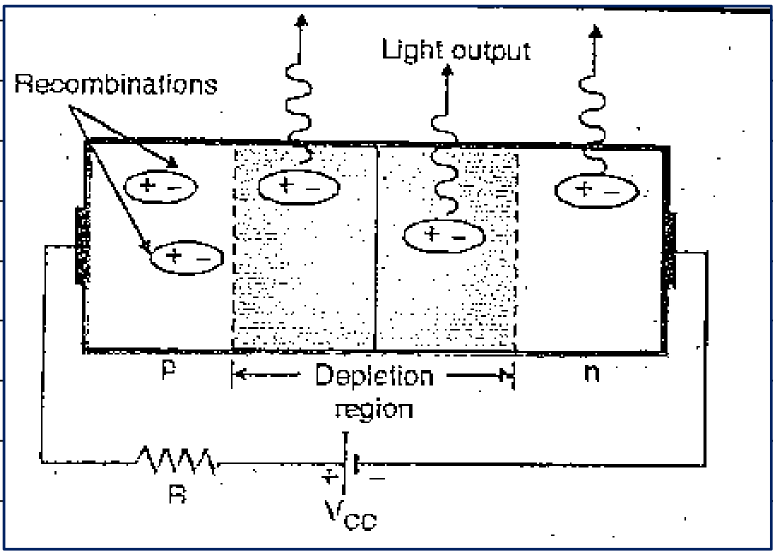
Without Biasing:

- When no voltage is applied, a depletion region is formed at the junction where no charge carriers are present.
- A barrier potential exists, preventing charge flow.

Forward Biased Condition:

- When a sufficient voltage is applied, the depletion region disappears.
- Electrons move from n-type to p-type, and holes move from p-type to n-type.
- When an electron recombines with a hole, energy is released in the form of photons, producing light emission.

Diagram of LED Structure and Working



(diagram showing the LED structure and recombination process.)

Characteristics of LED

Voltage-Current (V-I) Characteristics:

- LED follows a non-linear V-I characteristic.

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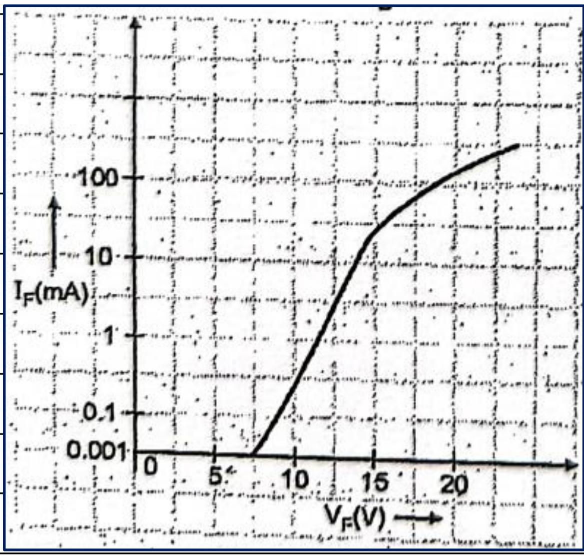
- It requires a minimum forward voltage (Threshold voltage) to start conduction.
- No significant current flows when reverse biased.



(V-I characteristic graph with Voltage (V) on the x-axis and Current (I) on the y-axis.)

Light Output Characteristics:

- The intensity of emitted light is directly proportional to the forward current.
- Light output depends on the recombination rate of charge carriers.

Power and Efficiency:

- The power output of an LED ranges between 200 to 500 μW.
- LEDs have high efficiency compared to incandescent bulbs.

Materials Used in LEDs and Their Wavelengths

- The emission wavelength of an LED depends on the energy band gap of the semiconductor material.
- Different materials used for LEDs are:
  - Gallium Arsenide (GaAs): Emits infrared light (~850-940 nm).
  - Gallium Phosphide (GaP): Emits red, yellow, and green light (550-700 nm).
  - Gallium Nitride (GaN): Emits blue and ultraviolet light (400-500 nm).

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- Aluminum Gallium Indium Phosphide (AlGaInP): Emits orange and red light (~590-630 nm).

Properties of LED for Optical Fiber Communication

- High Speed: LED responds quickly, making it suitable for high-speed data transmission.
- Monochromatic Light: Emits narrowband light, reducing signal dispersion in optical fibers.
- Energy Efficient: LEDs consume low power and generate less heat.
- Compact Size: Small and lightweight, making them ideal for fiber-optic communication systems.
- Long Lifespan: LEDs have a long operational life compared to traditional light sources.
- Reliability: LEDs are highly durable, resistant to vibrations and temperature variations.

-> c) Semiconductor Materials Used for Various Optical Sources

- Semiconductor materials play a crucial role in optical sources like LEDs, laser diodes, and photodetectors. These materials determine the efficiency, wavelength, and overall performance of the optical devices.
- Different semiconductor materials are chosen based on their band-gap energy, light emission properties, and application requirements.

Common Semiconductor Materials for Optical Sources:

1. Gallium Arsenide (GaAs):

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- GaAs is a direct band-gap semiconductor widely used in light-emitting diodes (LEDs) and laser diodes. Its high electron mobility and efficient radiative recombination make it suitable for infrared (IR) and visible light emission.
- It is commonly used in fiber optic communication, infrared remote controls, and optical data transmission applications.
- GaAs-based lasers are also used in barcode scanners, optical storage devices, and medical applications.

2. Indium Phosphide (InP):
- InP is another direct band-gap semiconductor used in optical communication systems, particularly in fiber-optic networks.
- It efficiently emits light in the infrared region, making it ideal for long-distance communication.
- InP-based lasers are used in high-speed optical communication and photonic integrated circuits.

3. Gallium Nitride (GaN):
- GaN is widely used in blue and ultraviolet (UV) LEDs and laser diodes.
- It has a wide band gap, allowing it to emit high-energy photons suitable for high-brightness applications.
- GaN-based LEDs are used in display screens, lighting, and high-density optical storage devices like Blu-ray discs.

4. Silicon (Si):
- Silicon is an indirect band-gap material, which makes it inefficient for light emission. However, it is used in photodetectors and optical sensors.
- Si-based photodetectors are used in cameras, solar cells, and optical receivers in fiber optic communication.
- Researchers are exploring silicon photonics to integrate optical components with electronic circuits for high-speed data transmission.

5. Germanium (Ge):

- Germanium is another indirect band-gap semiconductor primarily used in photodetectors and infrared applications.
- It is used in optical fiber detectors, night vision devices, and infrared cameras.
- Ge is often combined with silicon to enhance the performance of photodetectors.

6. Aluminum Gallium Arsenide (AlGaAs):

- AlGaAs is a compound semiconductor used in laser diodes and high-efficiency LEDs.
- It allows tunable emission wavelengths, making it suitable for various optoelectronic applications.
- It is used in optical storage, medical applications, and industrial laser systems.

7. Zinc Selenide (ZnSe):

- ZnSe is used for mid-infrared lasers and optical coatings.
- It is an essential material for CO2 laser optics and infrared imaging systems.
- ZnSe-based devices are used in scientific research and medical applications.

Factors Affecting the Selection of Semiconductor Materials for Optical Sources:

- Band Gap Energy: Determines the emission wavelength and efficiency of light emission.
- Carrier Mobility: Higher mobility allows for faster electron transitions, improving device performance.
- Thermal Stability: Materials should withstand temperature variations without degrading performance.

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- Compatibility with Fabrication Technology: The material should be compatible with existing semiconductor manufacturing processes for large-scale production.

These semiconductor materials form the foundation of modern optical communication, lighting, and sensing technologies, driving advancements in electronics and photonics.

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-> a) - A semiconductor laser diode operates based on three fundamental processes: absorption, spontaneous emission, and stimulated emission.

- These processes determine the working of lasers and their ability to generate coherent light.
- Below is a detailed explanation of these processes along with a diagram space for better understanding.

Absorption :

- In a semiconductor material, electrons naturally tend to stay at the lower energy level, which is called the ground state (E1).
- When a photon of specific energy strikes an electron in this state, the electron absorbs the energy.
- This increases the total energy contained by the electron, causing it to jump to a higher energy level called excited state (E2). This process is known as absorption.

- If the photon's energy is equal to or greater than (E2 - E1), the electron transitions completely to the higher energy level.
- If the photon's energy is insufficient, the electron may shift to an intermediate metastable state instead of reaching E2.
- After reaching the higher energy level, the electron remains in this excited state for a short time before returning to its lower energy level.

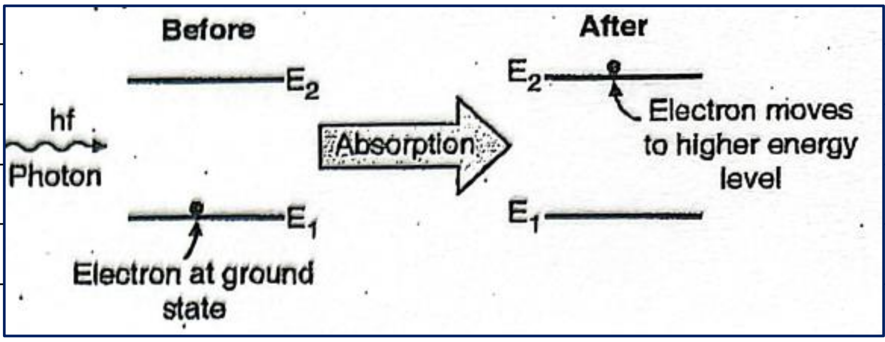

Diagram: Absorption process showing electron transition from E1 to E2 due to photon energy.

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Spontaneous Emission :

- After an electron absorbs energy and moves to the higher energy level, it does not stay there permanently.
- The electron will eventually return to its ground state without external influence, releasing the excess energy in the form of a photon.
- This process is called spontaneous emission.

- The emitted photon carries an energy equal to (E2 - E1).
- The direction, phase, and polarization of these emitted photons are random, meaning they do not contribute to a coherent light source.
- This process occurs naturally within a very short time interval (nanoseconds).

Spontaneous emission is responsible for the initial light output in semiconductor lasers, but it does not result in a coherent or amplified light source like in lasers.

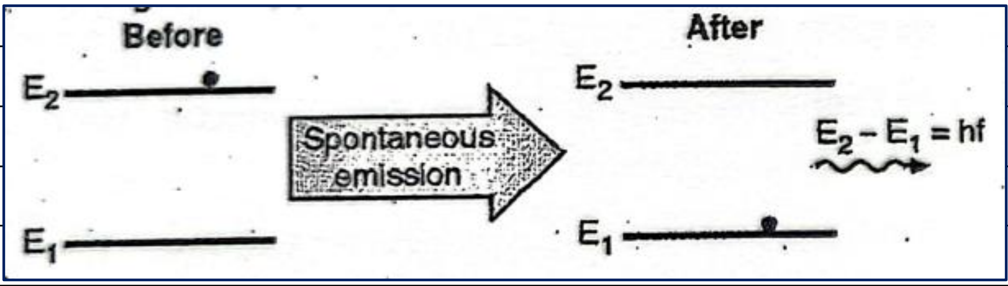

Diagram: Electron returning to ground state, releasing a photon randomly.

Stimulated Emission :

- Stimulated emission is the key process behind laser operation.
- In this case, an external photon interacts with an electron that is already in the excited state (E2).
- This photon stimulates the electron to drop back to the ground state (E1) while releasing another photon in the process.
- The newly emitted photon is identical to the original photon in terms of energy, phase, polarization, and direction.

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- This leads to the amplification of light, as two identical photons are generated from one interaction.
- This process continues in a controlled environment, leading to coherent, monochromatic laser light.
- The gain medium in the semiconductor laser is designed to maximize stimulated emission, ensuring that more photons are generated than absorbed.

Stimulated emission is the primary principle behind laser action, as it results in the emission of coherent and intense light.

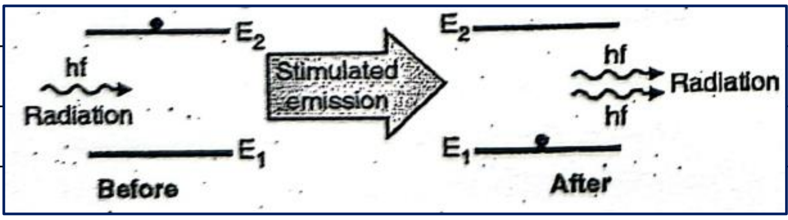

Diagram: Illustration of stimulated emission showing photon interaction and generation of two identical photons.

-> b) Comparison of LED and LASER

Basis of Difference	LED (Light Emitting Diode)	LASER (Light Amplification by Stimulated Emission of Radiation)
Definition	A semiconductor device that emits light when current flows through it.	A device that amplifies light through stimulated emission of radiation.
Working Principle	Based on **electroluminescence** (spontaneous emission of photons).	Based on **stimulated emission** (photon triggers more photon emission).
Chromaticity	**Polychromatic** (produces multiple wavelengths).	**Monochromatic** (produces a single wavelength).
Coherence	**Non-coherent** light (out of phase photons).	**Coherent** light (in-phase photons).
Directionality	Light spreads in multiple directions (**divergent beam**).	Light travels in a single direction (**highly directional beam**).
Optical Power Output	Low optical power output.	High optical power output.
Applications	Used in **LED displays, traffic lights, indicator lights, and automobile headlamps.**	Used in **fiber optics, optical communication, medical surgery, and metal cutting.**

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-> c)

- A Gallium Arsenide (GaAs) laser is a type of semiconductor laser that operates based on stimulated emission of radiation.
- It is widely used in optical communication, barcode scanners, and medical applications due to its efficiency and small size.
- The given laser operates at a wavelength of 900 nm with a cavity length of 300 micrometers and a refractive index of 4.3.

Frequency Spacing (Δf) :

- Frequency spacing represents the difference in frequency between adjacent modes inside the laser cavity.
- It is calculated using the formula:

$$\Delta f = c / (2 \times n \times L)$$

where:
- c = Speed of light = 3×10^8 m/s
- n = Refractive index = 4.3
- L = Cavity length = 300×10^{-6} m

- Substituting the values:

$$\Delta f = (3 \times 10^8) / (2 \times 4.3 \times 300 \times 10^{-6})$$
$$\Delta f = (3 \times 10^8) / (2.58 \times 10^{-3})$$
$$\Delta f \approx 1.16 \times 10^{11} \text{ Hz (116 GHz)}$$

- Thus, the frequency spacing of the laser is 116 GHz.

Wavelength Spacing ($\Delta \lambda$) :

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- Wavelength spacing represents the difference in wavelength between adjacent longitudinal modes.
- It is determined using the relation:

$$\Delta\lambda = \lambda^2 / (2 \times n \times L)$$

where:

- λ = Wavelength = 900 nm = 900×10^{-9} m
- n = Refractive index = 4.3
- L = Cavity length = 300×10^{-6} m

- Substituting the values:

$$\Delta\lambda = (900 \times 10^{-9})^2 / (2 \times 4.3 \times 300 \times 10^{-6})$$

$$\Delta\lambda = (8.1 \times 10^{-13}) / (2.58 \times 10^{-3})$$

$$\Delta\lambda \approx 3.14 \times 10^{-4}\ \mu m\ (0.314\ nm)$$

- Thus, the wavelength spacing of the laser is 0.314 nm.

Number of Modes (N) :

- The number of longitudinal modes supported by the laser cavity is given by:

$$N = \lambda / \Delta\lambda$$

where:

- λ = Wavelength = 900×10^{-9} m
- $\Delta\lambda$ = Wavelength spacing = 3.14×10^{-9} m

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- Substituting the values:

$N = (900 \times 10^{-9}) / (3.14 \times 10^{-9})$

$N \approx 286$

- Thus, the number of modes inside the laser cavity is approximately 286.